

Introduction to Artificial Intelligence

Week 14

Agenda

- Generative AI

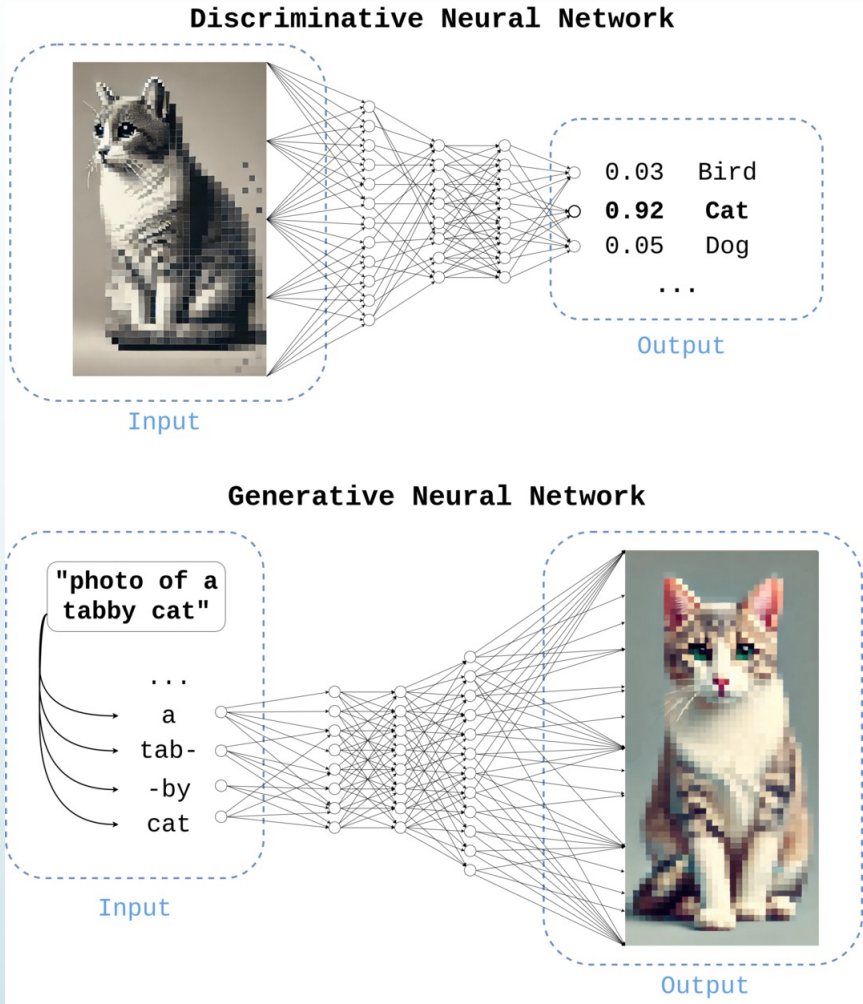
Introduction to Generative AI

Generative AI History

- 1957: Frank Rosenblatt's Perceptron
- 1961: the first historical example of GenAI ELIZA by Joseph Weizenbaum
- 1970s: backpropagation was proposed
- Till 2013th mostly *discriminative* were applied due to the difficulty of *generative* models
- 2013: VAE by Diederik P. Kingma and Max Welling
- 2014: GAN by Ian Goodfellow
- 2015: Diffusion Models introduced by Sohl-Dicktein et al.
- 2017: Transformer network by Google researchers
- 2018: OpenAI intoruduced GPT-1 (Generative Pre-trained Transformer)
- 2020s: GenAI Boom
- 2022: public release of ChatGPT for general-purpose text-based tasks

Machine Learning

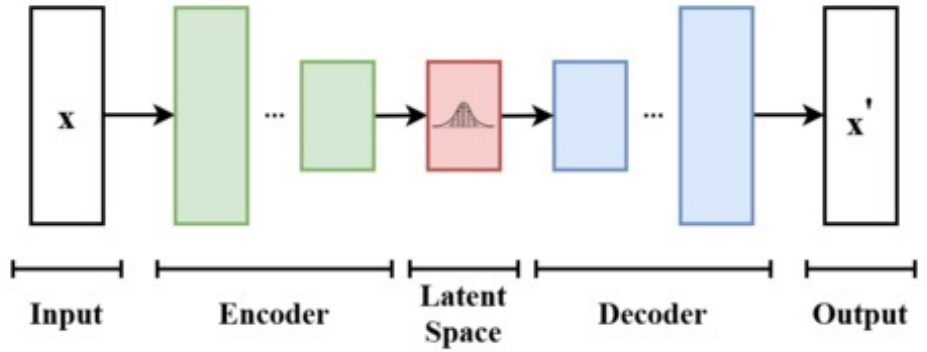
- *Machine Learning* (ML) is a field of study in AI concerned with the development and study of statistical algorithms that can learn from data and generalise to unseen data, and thus perform tasks without explicit instructions
- *Deep Learning* (DL) is a part of ML
- DL focuses on utilizing multilayered neural networks to perform tasks such as classification, regression, and representation learning
- ML uses both *discriminative* models and *generative* models to predict data



Generative AI Types

- Variational Autoencoders (VAE)
- Generative Adversarial Networks (GAN)
- Diffusion Models (DM)
- Large Language Models (LLM)
- …

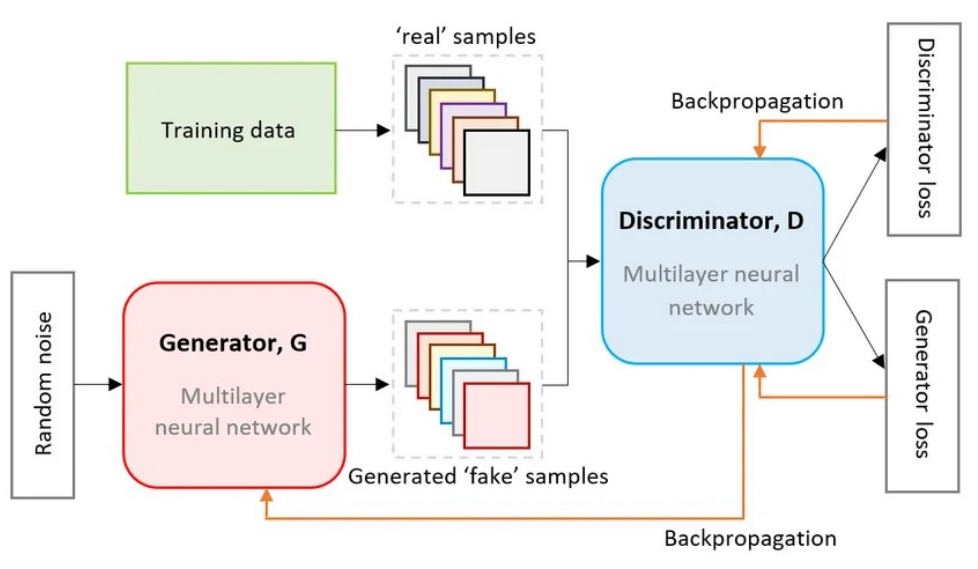
VAE



- Variational autoencoder is an ANN architecture introduced by Diederik P. Kingma and Max Welling in 2013
- *Encoder* transforms input data into parameter distributions
- *Decoder* takes a sampled vector from the latent distribution and reconstructs the data
- Due to random sampling, even for the same input data the result will vary, which provides variety in generation
- Considered to be a bridge between *Classic Autoencoders* and GAN
- Its strength lies in its ability to model data distributions rather than simply copying examples

GAN

- The GAN concept was initially developed by Ian Goodfellow and his colleagues in June 2014
- In a GAN, two neural networks, the *generator* and the *discriminator*, compete with each other in the form of a zero-sum game, where one agent's gain is another agent's loss
- The generator creates *synthetic data* by transforming random noise into samples that resemble the training dataset
- The discriminator is trained to distinguish the *authentic data* from synthetic data produced by the generator



Diffusion Models

- Diffusion Models were introduced by Sohl-Dicktein et al
- DM creates data by sequentially reconstructing structure from random noise. Unlike GANs and VAEs, DMs operate on a "chaos → order" principle.

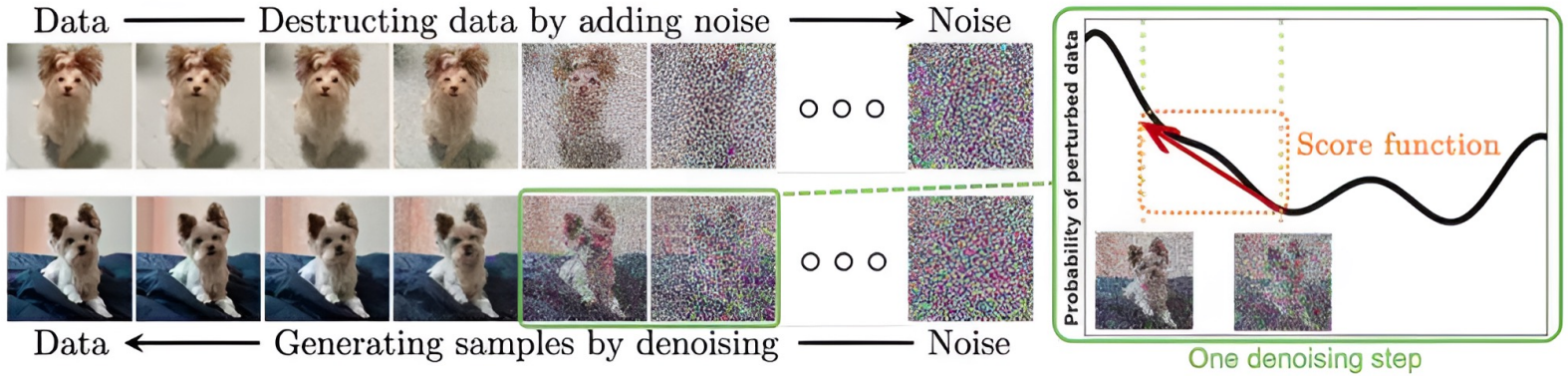
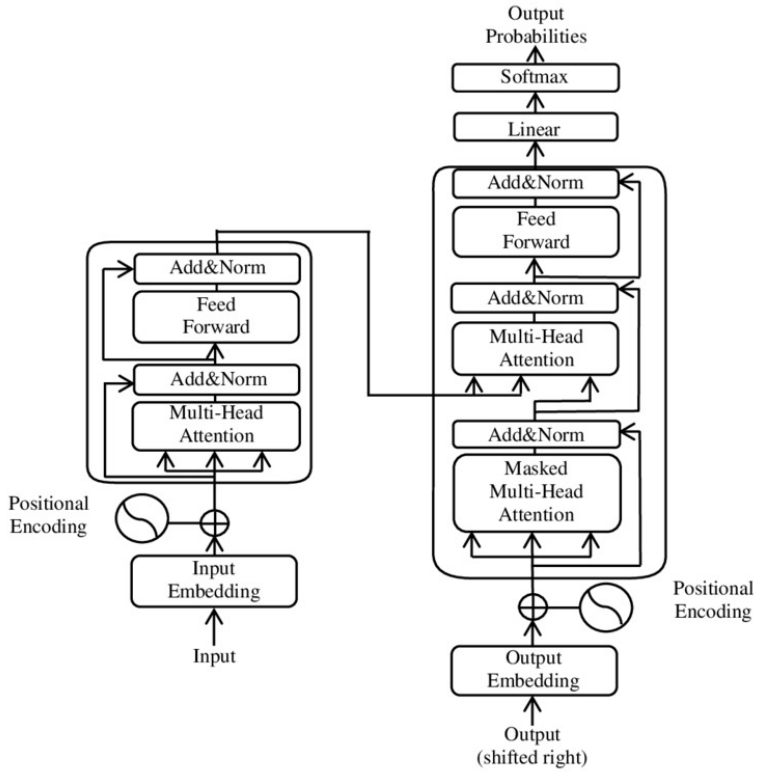


Fig. 2. Diffusion models smoothly perturb data by adding noise, then reverse this process to generate new data from noise. Each denoising step in the reverse process typically requires estimating the score function (see the illustrative figure on the right), which is a gradient pointing to the directions of data with higher likelihood and less noise.

LLM

- LLM is a language model trained with self-supervised ML on a vast amount of text, designed for natural language processing tasks, especially language generation
- They consist of billions to trillions of parameters and operate as general-purpose sequence models, generating, summarizing, translating, and reasoning over text
- Since 2023, many LLMs have been trained to be *multimodal*, having the ability to also process or generate other types of data, such as images, audio, or 3D mesh. These LLMs are also called large multimodal models (LMMs)



Applications

- Text
  - creating articles, scripts, marketing materials, translations
- Images
  - design, advertising, concept art, fashion
- Sound and music
  - speech synthesis, melodic generation, voiceover
- Video
  - creating animated avatars, special effects, editing
- Programming
  - code completion
- Science
  - molecular modeling, protein structure prediction
- Games
  - generating textures, locations, and dialogue.

Advantages

- Decreases the costs
- Increases effectiveness and efficiency
- Expanding creative possibilities
- Accessibility and democratization of technology
- …

Issues

- Quality and reliability of generated content
  - Hallucinations
- Ethical issues
  - Deepfakes
- Legal uncertainties
- Bias and discrimination
- Technical and resource limitations
- Overtraining and generalization
- Safety and Abuse
- Verification and trust issues
  - Watermarks?
- Human substitution

Summary

- Generative AI

Next term: ML